OCR A-Level Further Mathematics (H245) — Core Pure Section 1.2

1.2: Matrices

1. What Is a Matrix?

A **matrix** is a rectangular array of numbers arranged in rows and columns. A matrix with m rows and n columns has **order** $m \times n$.

We write $\mathbf{A} = (a_{ij})$ where a_{ij} is the entry in row i, column j. A **square matrix** has $m = n$. A **column vector** is $n \times 1$; a **row vector** is $1 \times n$.

2. Matrix Arithmetic

Addition and Scalar Multiplication

Two matrices of the same order are added element-wise. Scalar multiplication multiplies every entry.

$$(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij}, \quad (k\mathbf{A})_{ij} = k\, a_{ij}$$

Matrix Multiplication

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times p$, then the product $\mathbf{AB}$ is $m \times p$ with entries

$$(\mathbf{AB})_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

The inner dimensions must match.

Matrix multiplication is **not commutative**: in general $\mathbf{AB} \neq \mathbf{BA}$. It *is* associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}.$$

Identity and Zero Matrices

The $n \times n$ identity matrix $\mathbf{I}$ has ones on the main diagonal and zeros elsewhere, satisfying $\mathbf{AI} = \mathbf{IA} = \mathbf{A}$. The zero matrix $\mathbf{0}$ has all entries zero.

3. Determinants

2×2 Determinant

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

3×3 Determinant (Expansion Along the First Row)

$$\det \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

Signs alternate $(+, -, +, \ldots)$ across the row. Expansion along any row or column gives the same answer — choose a row/column with zeros to make life easier.

Key Properties

- $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$
- $\det(\mathbf{A}^T) = \det(\mathbf{A})$
- $\det(k\mathbf{A}) = k^n \det(\mathbf{A})$ for an $n \times n$ matrix
- $\det(\mathbf{A}^{-1}) = 1/\det(\mathbf{A})$
- If $\det(\mathbf{A}) = 0$, $\mathbf{A}$ is **singular** (no inverse exists).

Geometric Meaning

$|\det(\mathbf{A})|$ is the scale factor for areas (2D) or volumes (3D) under the linear transformation $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$. Negative determinant signifies a reflection (reverses orientation).

4. Inverse Matrices

2×2 Inverse

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \implies \mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \text{ (provided } ad - bc \neq 0\text{).}$$

3×3 Inverse (Adjugate Formula)

For a 3×3 matrix $\mathbf{A}$ with $\det \mathbf{A} \neq 0$:

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} \operatorname{adj}(\mathbf{A})$$

The adjugate (adjoint) is the *transpose of the cofactor matrix*:

1. Compute the cofactor $C_{ij} = (-1)^{i+j} M_{ij}$, where M_{ij} is the determinant of the 2×2 minor obtained by deleting row i and column j.
2. Form the cofactor matrix $\mathbf{C} = (C_{ij})$.
3. Transpose: $\operatorname{adj}(\mathbf{A}) = \mathbf{C}^T$.

Inverse Properties

- $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ — *note the reversed order.*
- $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$, $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$.

5. Solving Simultaneous Equations with Matrices

A system $a_{11}x + a_{12}y + a_{13}z = d_1$, etc., becomes $\mathbf{A}\mathbf{x} = \mathbf{d}$, with solution

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{d} \text{ (when } \mathbf{A} \text{ is invertible).}$$

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Geometric Interpretation of 3×3 Systems

$\det \mathbf{A}$	Geometry	Number of solutions
$\neq 0$	Three planes meet at a single point	Unique solution
$= 0$, consistent	Three planes share a line (sheaf) or coincide	Infinitely many solutions
$= 0$, inconsistent	Three planes form a prism (no common point)	No solution

OCR Exam Tip: When $\det \mathbf{A} = 0$, you must investigate further by elimination to decide between the "sheaf" and "prism" cases.

6. Linear Transformations

A 2×2 (or 3×3) matrix $\mathbf{M}$ represents a linear transformation $\mathbf{x} \mapsto \mathbf{M}\mathbf{x}$. Linear transformations fix the origin and preserve straight lines.

Standard 2D Transformations

Transformation	Matrix
Rotation by θ anticlockwise	$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$
Reflection in $y = x\tan\theta$	$\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$
Reflection in x-axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Reflection in y-axis	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$
Enlargement, scale factor k	$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$

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Stretch parallel to x-axis, factor k	$\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$
Shear parallel to x-axis by k	$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$

Composing Transformations

If you first apply $\mathbf{P}$ then $\mathbf{Q}$, the combined matrix is $\mathbf{QP}$ — the *second* transformation is on the *left*.

Find the matrix for a rotation 90° anticlockwise followed by reflection in the x-axis.
$\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $\mathbf{F} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.
Combined: $\mathbf{FR} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$.
This is reflection in the line $y = -x$.

7. Invariant Points and Lines

Invariant point: a point $\mathbf{p}$ with $\mathbf{Mp} = \mathbf{p}$.
Invariant line: a line mapped to itself (as a set of points — individual points may move along the line).
Line of invariant points: a line on which every point is invariant.

Solve $(\mathbf{M} - \mathbf{I})\mathbf{p} = \mathbf{0}$ to find invariant points. For invariant lines, set up a generic point on the line, transform it, and require the image to lie on the line.

8. Eigenvalues and Eigenvectors

A non-zero vector $\mathbf{v}$ is an **eigenvector** of $\mathbf{A}$ with **eigenvalue** λ if

$$\mathbf{Av} = \lambda\mathbf{v}.$$

Finding Eigenvalues — the Characteristic Equation

$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \Leftrightarrow (\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ with $\mathbf{v} \neq \mathbf{0}$, which requires:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

For a 2×2 matrix this is a quadratic in λ; for 3×3 it is a cubic.

Finding Eigenvectors

For each eigenvalue λ, solve $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ — you always get an infinite family (the solution is a line/plane). Pick a convenient non-zero representative.

Find eigenvalues and eigenvectors of $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

$\det(\mathbf{A} - \lambda\mathbf{I}) = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 = 0.$

Eigenvalues: $\lambda = 2, 5$.

$\lambda = 5$: $\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}\mathbf{v} = 0 \Rightarrow \mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$\lambda = 2$: $\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}\mathbf{v} = 0 \Rightarrow \mathbf{v} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

9. Diagonalisation

If an $n \times n$ matrix $\mathbf{A}$ has n linearly independent eigenvectors, it can be written as

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$$

where $\mathbf{D}$ is diagonal with eigenvalues on the diagonal, and $\mathbf{P}$ has the corresponding eigenvectors as columns (in the same order).

Why It Matters

Powers and exponentials become easy:

$$\mathbf{A}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$$

where $\mathbf{D}^n$ just raises each diagonal entry to the nth power.

Using the previous example, compute $\mathbf{A}^{10}$.
$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}$, $\mathbf{D} = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$.
$\det \mathbf{P} = -3$, $\mathbf{P}^{-1} = -\frac{1}{3}\begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$.
$\mathbf{A}^{10} = \mathbf{P}\begin{pmatrix} 5^{10} & 0 \\ 0 & 2^{10} \end{pmatrix}\mathbf{P}^{-1}$.

OCR Exam Tip: Diagonalisation questions are usually structured: part (a) find eigenvalues and eigenvectors, part (b) hence find $\mathbf{A}^n$ or solve a coupled recurrence. Lay out $\mathbf{P}$, $\mathbf{D}$, $\mathbf{P}^{-1}$ clearly.

10. Cayley–Hamilton Theorem (Support)

Every square matrix satisfies its own characteristic equation. If $\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0$, then

$$\mathbf{A}^n + c_{n-1}\mathbf{A}^{n-1} + \cdots + c_0\mathbf{I} = \mathbf{0}.$$

Use this to express higher powers of $\mathbf{A}$ in terms of lower ones, or to find $\mathbf{A}^{-1}$.

11. Exam Style Strategy

- Write determinants out carefully — sign errors in cofactor expansions are the single most common mistake.
- For 3×3 systems, start with the determinant; only do elimination when $\det = 0$.
- For composite transformations, multiply in the order *last first* (the last transformation goes on the left).
- Always verify an eigenvector by computing $\mathbf{A}\mathbf{v}$ and checking it equals $\lambda\mathbf{v}$.

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